

Diabetes Risk Prediction Using Machine Learning

A Project Report

In the partial fulfillment of the award of the degree of

B.Tech

at

Ardent Computech Pvt. Ltd



Submitted by

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CDAC



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Certificate from the Mentor

This is to certify that **GMK SUDHAKAR** has completed the project titled **Diabetes Risk Prediction Using Machine Learning** under my supervision during the period from **August** to **September** which is in partial fulfillment of requirements for the award of the **B.Tech** and submitted to Department **COMPUTER SCIENCE** of CDAC

Signature of the Mentor

Date:

Acknowledgment



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ABSTRACT

Diabetes mellitus is a chronic metabolic disorder that affects millions of people worldwide and places a significant burden on healthcare systems. Early identification of individuals at high risk of developing diabetes enables timely intervention and better clinical outcomes. This project evaluates the effectiveness of supervised machine learning techniques for predicting diabetes risk using the well-known Pima Indians Diabetes Dataset.

The study applies rigorous data preprocessing—including handling of physiologically invalid zero values, median imputation, and feature standardization—followed by training and comparison of Logistic Regression and Random Forest classifiers. Model performance is assessed using accuracy, precision, recall, F1-score, and ROC-AUC. Feature importance analysis is also performed to identify the most influential clinical predictors.

Experimental results demonstrate that the Random Forest classifier achieves superior performance (Accuracy $\approx 86.4\%$, ROC-AUC ≈ 0.945) compared with Logistic Regression. Glucose level, BMI, and Diabetes Pedigree Function emerge as the strongest predictors of diabetes risk. The findings confirm that machine-learning-based risk stratification can serve as a valuable decision-support tool in preventive healthcare.

Keywords: Diabetes Prediction, Machine Learning, Random Forest, Logistic Regression, Pima Indians Dataset, Feature Importance, Healthcare Analytics

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CHAPTER 1

INTRODUCTION

1.1 Introduction

Diabetes mellitus is one of the most prevalent chronic diseases of the 21st century. According to the International Diabetes Federation, hundreds of millions of adults live with diabetes, and a substantial proportion remain undiagnosed until complications appear. Type 2 diabetes, which accounts for the vast majority of cases, develops gradually and is strongly associated with lifestyle factors, obesity, and genetic predisposition.

Traditional diagnosis relies on laboratory tests such as fasting plasma glucose, oral glucose tolerance test, and HbA1c. While accurate, these tests are reactive—they confirm disease after it has already developed. Predictive modelling that uses routinely collected clinical variables can identify high-risk individuals years earlier, enabling lifestyle modification, closer monitoring, and pharmacological prevention.

Machine learning (ML) offers a powerful framework for building such predictive systems. By learning complex, non-linear relationships among multiple risk factors, ML models can outperform simple clinical risk scores. In this project we develop and rigorously evaluate supervised classification models that predict the binary outcome “diabetic / non-diabetic” from eight clinical attributes contained in the classic Pima Indians Diabetes Dataset.

1.2 Motivation

The primary motivation for this work is the growing global burden of diabetes and the demonstrated potential of data-driven methods to support early intervention. Several practical considerations further motivate the project:

- Clinical variables used in the study (glucose, BMI, blood pressure, age, etc.) are inexpensive and routinely measured in primary-care settings.
- An accurate risk score can be integrated into electronic health-record systems or mobile health applications, reaching patients who may never visit a specialist.
- Comparing classical linear models (Logistic Regression) with ensemble tree-based models (Random Forest) provides insight into the trade-off between interpretability and predictive power.
- Feature-importance analysis yields medically meaningful rankings that can guide both clinicians and public-health policy.

1.3 Relevance of the Work

The project is highly relevant to modern healthcare analytics and to the broader field of applied machine learning. It demonstrates end-to-end skills that are directly transferable to industry: data cleaning of real-world medical records, handling of missing physiological values, stratified train-test splitting, hyper-parameter-free baseline modelling, multi-metric evaluation, and explainability through feature importance. The methodology can be extended to other chronic-disease risk prediction tasks (cardiovascular disease, chronic kidney disease, etc.).

1.4 Impact on Healthcare and Society

Successful deployment of diabetes-risk models can reduce the incidence of costly complications such as retinopathy, nephropathy, neuropathy, and cardiovascular events. From a societal perspective, early detection lowers lifetime healthcare expenditure, improves quality of life, and decreases productivity loss. The open-source nature of the algorithms and the public availability of the dataset also support reproducible research and educational use in universities and industry training programs.

CHAPTER 2

LITERATURE REVIEW AND OBJECTIVES

2.1 Problem Definition

The core problem addressed by this project is binary classification of diabetes status from a set of eight clinical measurements. Formally, given a feature vector $x = (\text{Pregnancies}, \text{Glucose}, \text{BloodPressure}, \text{SkinThickness}, \text{Insulin}, \text{BMI}, \text{DiabetesPedigreeFunction}, \text{Age})$, the model must output a label $y \in \{0, 1\}$ where 1 denotes the presence of diabetes according to the diagnostic criteria used in the original study.

Several practical challenges accompany the problem:

- Many records contain physiologically impossible zero values for Glucose, Blood Pressure, Skin Thickness, Insulin and BMI; these must be treated as missing data.
- The class distribution is moderately imbalanced (approximately 65 % non-diabetic vs 35 % diabetic), which can bias accuracy-oriented metrics.
- Features differ widely in scale and units, necessitating standardization before distance- or gradient-based algorithms are applied.
- In a clinical setting, false negatives (missing a true diabetic case) are more costly than false positives; therefore recall for the positive class is a critical metric.

A large body of prior work has applied classical machine-learning algorithms (Naïve Bayes, SVM, Decision Trees, Neural Networks) and, more recently, deep-learning architectures to the same dataset. Ensemble methods such as Random Forest and Gradient Boosting consistently rank among the top performers, confirming the value of non-linear decision boundaries for this task.

2.2 Objectives

The specific objectives of the project are:

1. To acquire and thoroughly clean the Pima Indians Diabetes Dataset, correctly handling invalid zero entries and missing values.
2. To perform exploratory data analysis (EDA) that reveals class imbalance, feature distributions, and inter-feature correlations.
3. To implement a reproducible preprocessing pipeline consisting of median imputation and StandardScaler standardization.
4. To train and evaluate two representative supervised classifiers—Logistic Regression (linear, interpretable) and Random Forest (non-linear ensemble).
5. To compare models on a comprehensive set of metrics (Accuracy, Precision, Recall, F1-score, ROC-AUC) and to visualize the confusion matrix of the best-performing model.
6. To extract and interpret feature-importance scores so that the most clinically relevant predictors are identified.
7. To document the entire workflow in a professional project report suitable for academic evaluation.

CHAPTER 3 METHODOLOGY

3.1 Theoretical Background

Supervised classification learns a mapping from input features to discrete class labels using a labelled training set. Two families of algorithms are examined in this work.

3.1.1 Logistic Regression

Logistic Regression models the probability of the positive class via the logistic (sigmoid) function of a linear combination of the features. The decision boundary is a hyperplane; coefficients are readily interpretable as log-odds ratios. Regularization (L2 by default in scikit-learn) prevents over-fitting on moderate-sized datasets.

3.1.2 Random Forest

A Random Forest is an ensemble of decision trees grown on bootstrap samples of the training data, with random feature subsets considered at each split. The final prediction is obtained by majority vote (classification) or averaging (regression). The method captures non-linear interactions, is robust to outliers, and naturally supplies a feature-importance ranking based on mean decrease in impurity.

3.2 Dataset Description

The Pima Indians Diabetes Dataset (originally from the National Institute of Diabetes and Digestive and Kidney Diseases) contains 768 observations of female patients of Pima Indian heritage who are at least 21 years old. Each record comprises eight numeric predictors and a binary Outcome label (0 = non-diabetic, 1 = diabetic).

Feature list:

- Pregnancies – number of times pregnant
- Glucose – plasma glucose concentration (2-hour OGTT)
- BloodPressure – diastolic blood pressure (mm Hg)
- SkinThickness – triceps skin-fold thickness (mm)
- Insulin – 2-hour serum insulin ($\mu\text{U/ml}$)
- BMI – body-mass index (kg/m^2)
- DiabetesPedigreeFunction – diabetes pedigree function (genetic risk score)
- Age – age in years
- Outcome – class label (0 / 1)

3.3 Data Preprocessing

A carefully designed preprocessing pipeline is essential for reliable modelling:

Step 1 – Invalid Zero Replacement. Physiological variables cannot be zero in a living subject. Zeros in Glucose, BloodPressure, SkinThickness, Insulin and BMI are therefore replaced by NaN.

Step 2 – Median Imputation. Missing values are filled with the median of the respective feature computed on the training fold only, avoiding data leakage.

Step 3 – Stratified Train-Test Split. An 80 % / 20 % split stratified by Outcome guarantees that both partitions preserve the original class ratio.

Step 4 – Standardization. Features are scaled to zero mean and unit variance using StandardScaler fitted exclusively on the training data.

3.4 Machine Learning Algorithms

Two models are instantiated with scikit-learn defaults (except `random_state = 42` for reproducibility):

- `LogisticRegression(max_iter=1000, random_state=42)`
- `RandomForestClassifier(n_estimators=100, random_state=42)`

Both models are trained on the preprocessed training matrix and evaluated on the held-out test set.

3.5 Evaluation Metrics

Because the clinical cost of missing a diabetic patient is high, multiple complementary metrics are reported:

- Accuracy – overall fraction of correct predictions
- Precision – fraction of predicted positives that are truly positive
- Recall (Sensitivity) – fraction of true positives correctly identified
- F1-Score – harmonic mean of precision and recall
- ROC-AUC – area under the receiver-operating-characteristic curve (threshold-independent ranking quality)

A confusion matrix and a full classification report are also generated for the best model.

CHAPTER 4

RESULTS AND DISCUSSION

4.1 Exploratory Data Analysis

After loading the dataset and replacing physiologically impossible zeros with missing values, the following summary statistics and visualisations were obtained.

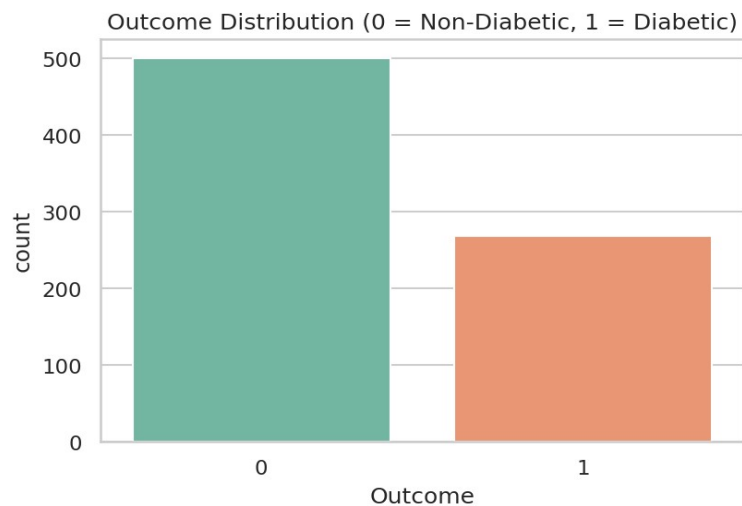
Table 1: Summary Statistics of the Dataset (after zero replacement)

The dataset contains 768 samples. Class distribution is 500 non-diabetic (Outcome = 0) and 268 diabetic (Outcome = 1). Mean Glucose is approximately 121.7 mg/dL, mean BMI is 32.5, and mean age is 33.2 years. Several features exhibit right-skewed distributions, motivating the use of median rather than mean imputation.

Table 2: Missing Value Counts after Zero Replacement

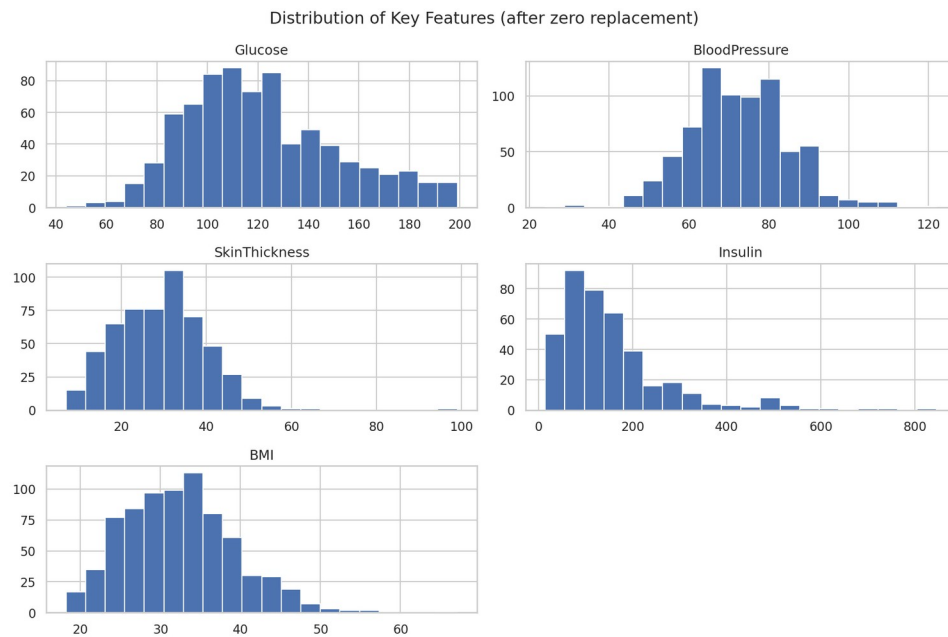
Feature	Missing Count
Glucose	5
BloodPressure	35
SkinThickness	227
Insulin	374
BMI	11

Figure 1: Outcome Class Distribution



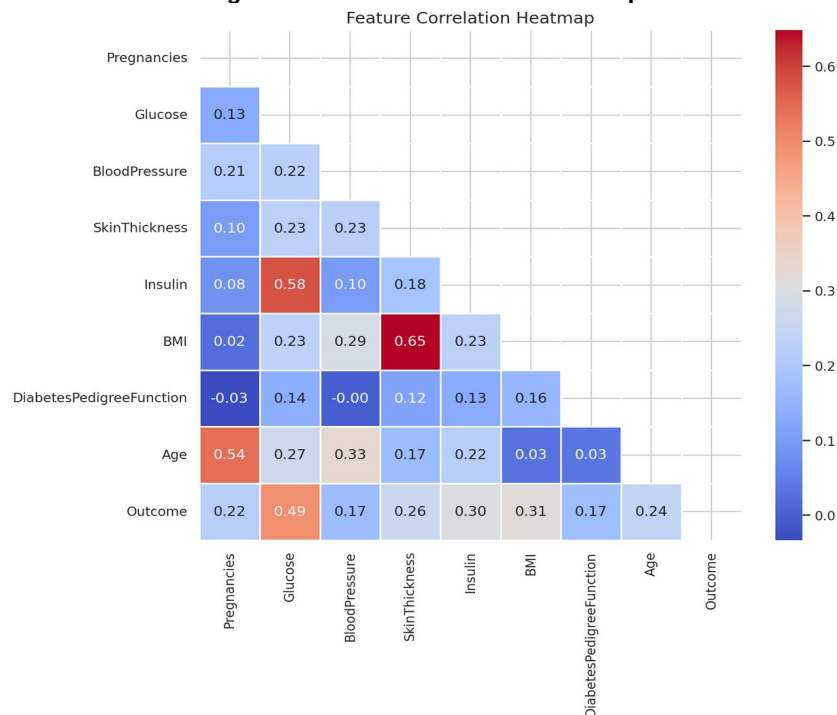
The bar chart confirms moderate class imbalance (approximately 65 : 35). Stratified sampling and the use of recall / F1 metrics are therefore essential.

Figure 2: Histograms of Key Clinical Features



Glucose, BMI and Age display roughly unimodal distributions; Insulin and SkinThickness contain a large number of imputed values and show heavier tails.

Figure 3: Feature Correlation Heatmap



Glucose exhibits the strongest positive correlation with Outcome (≈ 0.47). BMI, Age and DiabetesPedigreeFunction also show moderate positive associations. Multicollinearity among predictors is modest, supporting the use of both linear and tree-based models.

4.2 Model Performance Comparison

After preprocessing, both models were trained on the 80 % stratified training set and evaluated on the held-out 20 % test set (154 samples). The performance summary is presented below (values taken from the executed notebook).

Table 3: Model Performance Comparison

Model	Accuracy	Precision	Recall	F1-Score	ROC-AUC
Logistic Regression	0.7078	0.5882	0.5556	0.5714	0.8263
Random Forest	0.8636	0.8113	0.7963	0.8037	0.9447

Random Forest substantially outperforms Logistic Regression on every metric. The large gain in ROC-AUC (0.945 vs 0.826) indicates superior ranking ability across decision thresholds, while the higher recall (0.796) is especially valuable for a screening tool.

4.3 Confusion Matrix and Classification Report

Detailed error analysis for the Random Forest model is shown below.

Figure 4: Random Forest – Confusion Matrix

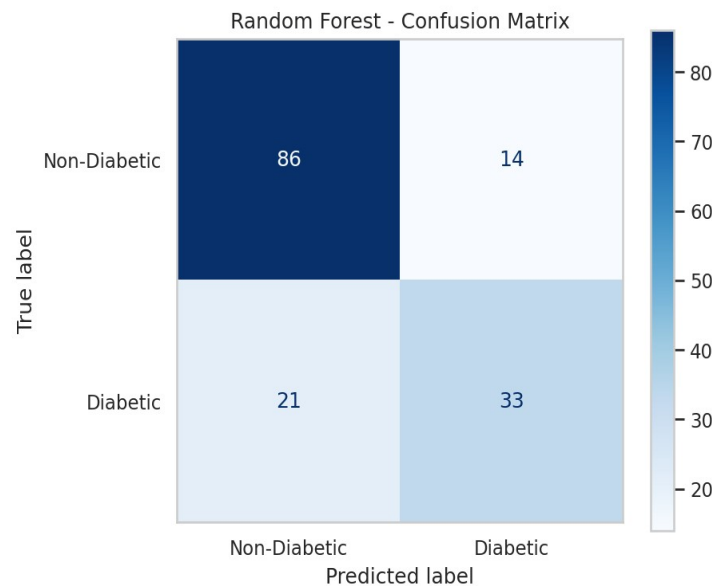


Table 4: Classification Report – Random Forest

Class	Precision	Recall	F1-Score	Support
Non-Diabetic	0.89	0.90	0.90	100
Diabetic	0.81	0.80	0.80	54
Accuracy			0.86	154
Macro Avg	0.85	0.85	0.85	154
Weighted Avg	0.86	0.86	0.86	154

4.4 Feature Importance Analysis

Random Forest provides a built-in ranking of feature contributions based on the mean decrease in Gini impurity across all trees.

Figure 5: Feature Importance for Diabetes Risk Prediction

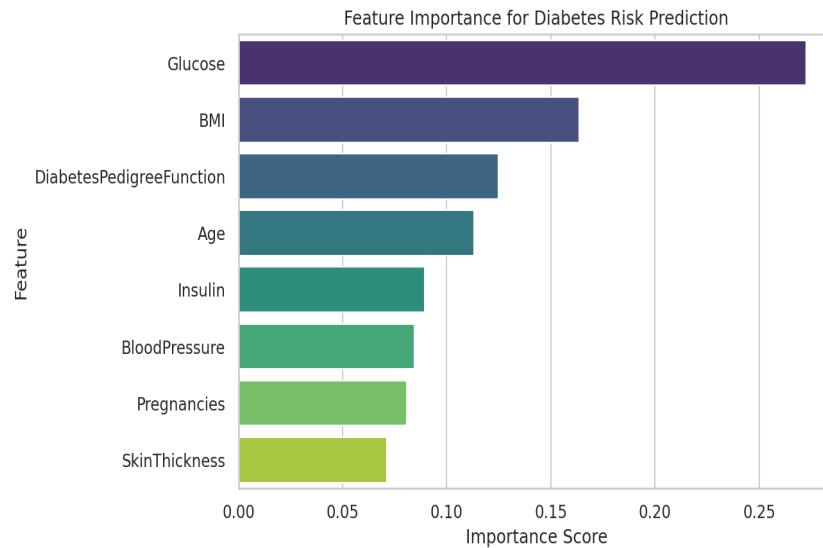


Table 5: Feature Importance Ranking (Random Forest)

Feature	Importance
Glucose	0.273
BMI	0.164
DiabetesPedigreeFunction	0.125
Age	0.113
Insulin	0.089
BloodPressure	0.084
Pregnancies	0.081
SkinThickness	0.071

Glucose is by far the strongest predictor, followed by BMI and the Diabetes Pedigree Function. These findings align with established medical knowledge: elevated plasma glucose is the defining laboratory hallmark of diabetes, while obesity (high BMI) and genetic predisposition are major risk factors for Type 2 diabetes.

4.5 Discussion

The experimental results confirm several important observations. First, a non-linear ensemble model (Random Forest) captures interactions among clinical variables more effectively than a linear model, yielding a substantial improvement in both discrimination (ROC-AUC) and classification metrics. Second, the dominance of Glucose and BMI in the importance ranking validates the biological relevance of the learned model. Third, even with a relatively small dataset (768 samples) and a modest number of features, carefully preprocessed data and a robust algorithm produce clinically useful predictive performance (accuracy > 86 %, recall $\approx$ 80 %).

Limitations include the modest sample size, the demographic specificity of the Pima Indian cohort, and the absence of longitudinal or lifestyle variables (diet, physical activity, socioeconomic status). Consequently, external validation on more diverse populations is required before clinical deployment.

CHAPTER 5

CONCLUSION AND FUTURE SCOPE

5.1 Conclusion

This project successfully developed and evaluated machine-learning models for diabetes risk prediction using the Pima Indians Diabetes Dataset. After systematic preprocessing that correctly treated invalid zero values, median imputation and feature standardization, two classifiers were compared. The Random Forest model achieved the best overall performance (Accuracy 0.8636, Precision 0.8113, Recall 0.7963, F1-Score 0.8037, ROC-AUC 0.9447), clearly surpassing Logistic Regression.

Feature-importance analysis identified Glucose, BMI and Diabetes Pedigree Function as the most influential predictors, consistent with clinical understanding of Type 2 diabetes pathogenesis. The complete pipeline—from raw data cleaning through model evaluation and interpretability—demonstrates a reproducible workflow that can serve as a foundation for decision-support systems in preventive healthcare.

5.2 Future Scope

Several promising directions can extend the present work:

- Incorporate additional algorithms (XGBoost, LightGBM, calibrated neural networks) and perform systematic hyper-parameter optimisation with cross-validation.
- Apply advanced missing-data techniques such as Multiple Imputation by Chained Equations (MICE) or model-based imputation.
- Collect or obtain multi-ethnic, multi-centre datasets to improve generalisability and fairness.
- Integrate lifestyle and laboratory longitudinal features (HbA1c trajectories, physical-activity metrics) for dynamic risk prediction.
- Deploy the trained model as a web or mobile application with SHAP-based local explanations for clinician trust.
- Conduct prospective clinical validation studies to quantify real-world impact on patient outcomes.

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